

Task 01:

Question 1:

The `waitpid(-1, &status, WNOHANG)` gives the following results for the given case:

- A child process terminates: +ve value which is the PID of the terminating child process.
- No child process terminates: 0
- There are more child processes: -1

Question 2:

If the parent would never call `wait()` or `waitpid()` then the order of termination of the two processes would be random and if the parent terminates first we would have a zombie process in the child process.

Question 3:

The argument of `waitpid()` is set to -1 to convey to the system that it is to wait for the termination of any of the child processes and not for the termination of a specific child process (case where the first argument is the process id of the specific child process).

Question 4:

Because of the line:

```
> exit((i + 1) % 256)
```

The exit code will just loop around.

With further experimentation we can report that even without the formula the exit code loops around with a mod of 256.